

Lesson 12 Demo 01

Evaluating Azure Kubernetes Service (AKS)

Objective: To evaluate Azure Kubernetes Service (AKS) for running multitier applications and test its deployment and scaling capabilities to simplify management overhead

Tools required: Azure account with administrator access

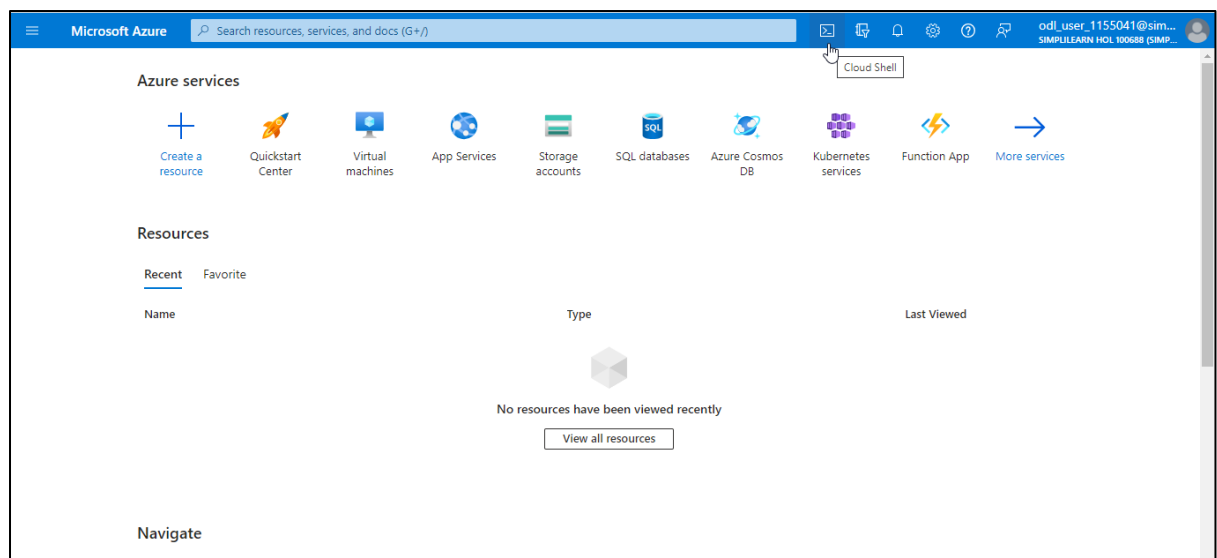
Prerequisites: None

Steps to be followed:

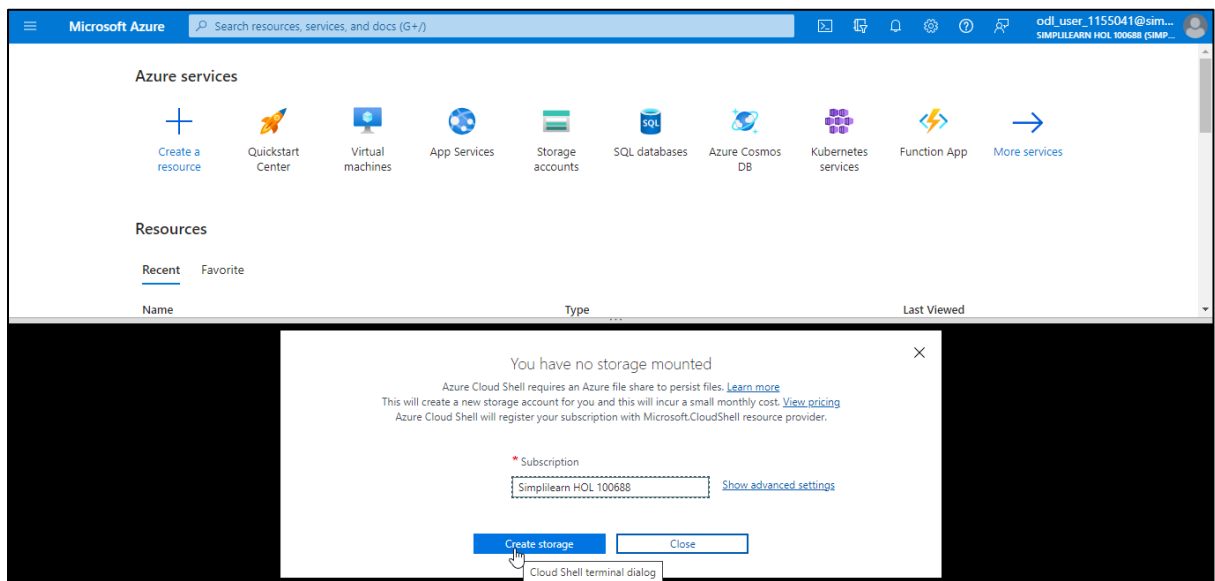
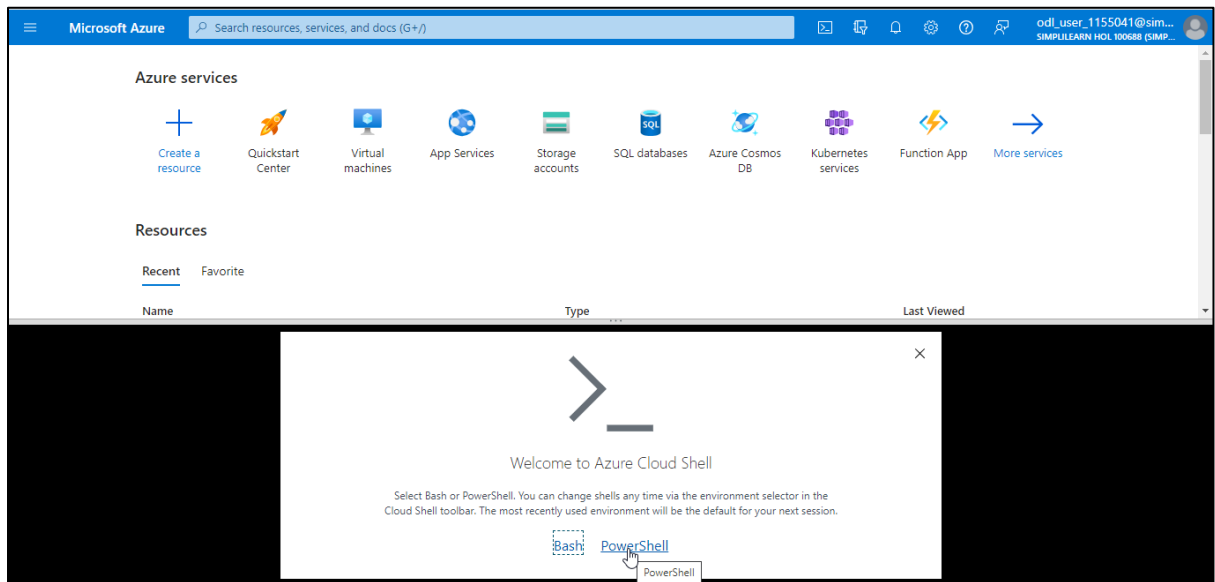
1. Register the Microsoft Kubernetes resource providers
2. Deploy an Azure Kubernetes Service (AKS) cluster
3. Deploy pods into the Azure Kubernetes Service (AKS) cluster
4. Scale containerized workloads in the Azure Kubernetes Service (AKS) cluster

Step 1: Register the Microsoft Kubernetes resource providers

1.1 Click on **Cloud Shell** after logging in to the Azure portal



1.2 Select PowerShell and click on Create storage



1.3 Run the following commands to register the **Microsoft.Kubernetes** and **Microsoft.KubernetesConfiguration** resource providers:

Register-AzResourceProvider -ProviderNamespace Microsoft.Kubernetes

Register-AzResourceProvider -ProviderNamespace

Microsoft.KubernetesConfiguration

```
PS /home/odl_user> Register-AzResourceProvider -ProviderNamespace Microsoft.Kubernetes

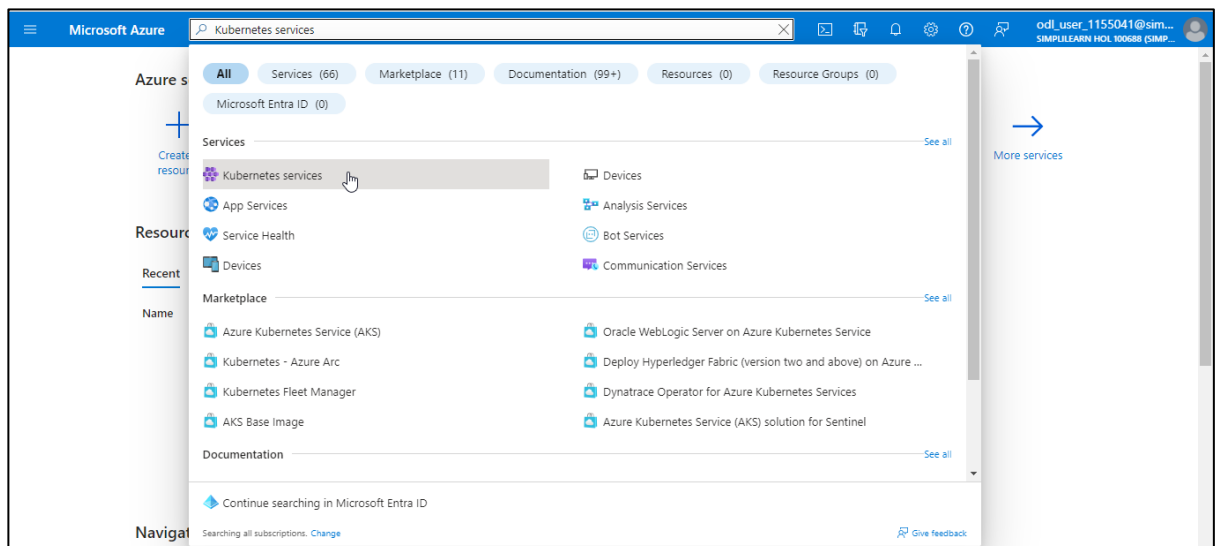
ProviderNamespace : Microsoft.Kubernetes
RegistrationState  : Registered
ResourceTypes     : {connectedClusters, locations, locations/operationStatuses, registeredSubscriptions...}
Locations         : {West Europe, East US, West Central US, South Central US...}
```

```
PS /home/odl_user> Register-AzResourceProvider -ProviderNamespace Microsoft.KubernetesConfiguration

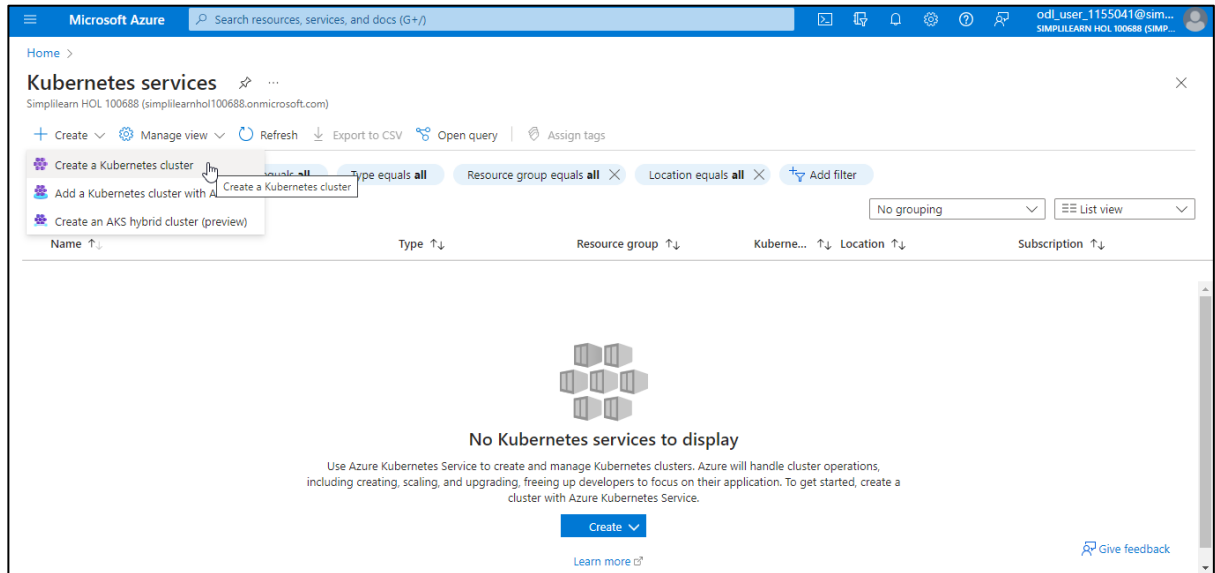
ProviderNamespace : Microsoft.KubernetesConfiguration
RegistrationState  : Registered
ResourceTypes     : {sourceControlConfigurations, extensions, fluxConfigurations, operations...}
Locations         : {East US, West Europe, West Central US, West US 2...}
```

Step 2: Deploy an Azure Kubernetes Service (AKS) cluster

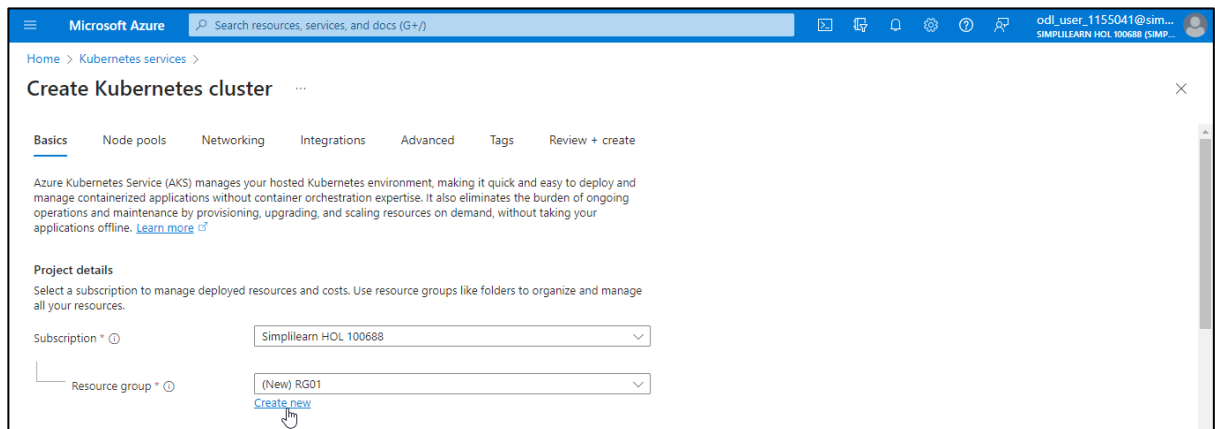
2.1 On the **Microsoft Azure** home page, search for and select **Kubernetes services**



2.2 In the **Kubernetes services** page, click on **+Create** and select **Create a Kubernetes cluster**



2.3 In the **Create Kubernetes cluster** page, navigate to the **Basics** tab. Now, click on **Create new** to create a new **Resource group (RG01)** under **Project details**.



2.4 Under **Cluster details**, provide the information as shown below:

Cluster details

Cluster preset configuration *

Dev/Test

To quickly customize your Kubernetes cluster, choose one of the preset configurations above. You can modify these configurations at any time.
[Compare presets](#)

Kubernetes cluster name * ⓘ

mykluster

Region * ⓘ

(US) East US

Availability zones ⓘ

Zones 1, 2, 3

AKS pricing tier ⓘ

Free

Kubernetes version * ⓘ

1.26.6

Automatic upgrade ⓘ

Enabled with patch (recommended)

Choose between local accounts or Azure AD for authentication and Azure RBAC or Kubernetes RBAC for your authorization needs.

Authentication and Authorization ⓘ

Local accounts with Kubernetes RBAC

2.5 Navigate to the **Node pools** tab

Microsoft Azure

Search resources, services, and docs (G+/I)

odl_user_1155041@sim...
SIMPLILEARN HOL 100688 (SIMP...)

Home > Kubernetes services >

Create Kubernetes cluster

Basics

Node pools

Networking

Integrations

Advanced

Tags

Review + create

Azure Kubernetes Service (AKS) manages your hosted Kubernetes environment, making it quick and easy to deploy and manage containerized applications without container orchestration expertise. It also eliminates the burden of ongoing operations and maintenance by provisioning, upgrading, and scaling resources on demand, without taking your applications offline. [Learn more](#)

Project details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Simplilearn HOL 100688

Resource group *

(New) RG01

[Create new](#)

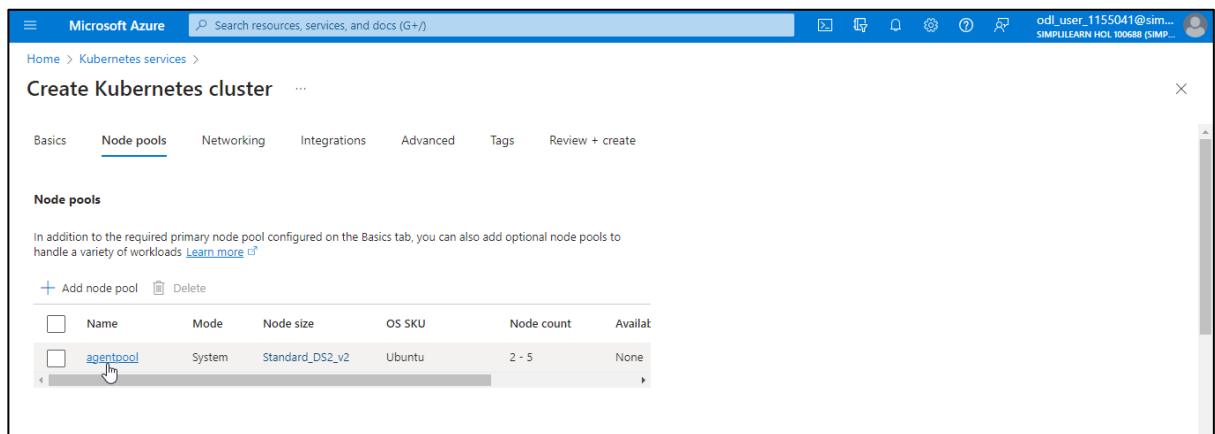
Cluster details

Cluster preset configuration *

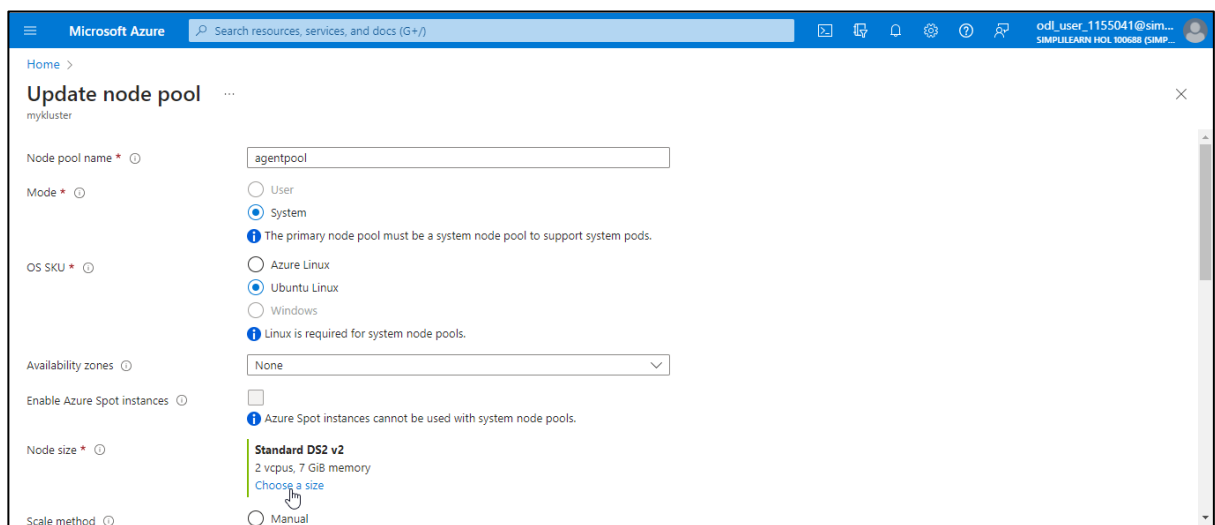
Dev/Test

To quickly customize your Kubernetes cluster, choose one of the preset configurations above. You can modify these configurations at any time.

2.6 Under the **Node pools**, click on **agentpool**



2.7 On the **Update node pool** page, provide the details as shown below and click on **Choose a size** under **Node size**



2.8 In the Select a VM size page, select D2s_v3

Microsoft Azure | Search resources, services, and docs (G+/I)

Home > Update node pool > Select a VM size

Search by VM size... | Display cost: Monthly | vCPUs: All | RAM (GiB): All | Add filter

Showing 792 VM sizes | Subscription: Simplilearn HOL 100688 | Region: East US | Learn more about VM sizes

Group by series

VM Size	Type	vCPUs	RAM (GiB)	Data disks	Max IOPS	Temp storage (GiB)	Premi
The latest generation M series sizes for extremely large databases or applications							
Most used by Azure users							
D2s_v3	General purpose	2	8	4	3200	16	Sup
D2as_v4	General purpose	2	8	4	3200	16	Sup
B2s	General purpose	2	4	4	1280	8	Sup
B2ms	General purpose	2	8	4	1920	16	Sup
DS2_v2	General purpose	2	7	8	6400	14	Sup
B4ms	General purpose	4	16	8	2880	32	Sup
D4s v3	General purpose	4	16	8	6400	32	Sup

2.9 Click Select

Microsoft Azure | Search resources, services, and docs (G+/I)

Home > Update node pool > Select a VM size

Search by VM size... | Display cost: Monthly | vCPUs: All | RAM (GiB): All | Add filter

Showing 792 VM sizes | Subscription: Simplilearn HOL 100688 | Region: East US | Current size: Standard_D2s_v3 | Learn more about VM sizes

Group by series

VM Size	Type	vCPUs	RAM (GiB)	Data disks	Max IOPS	Temp storage (GiB)	Premi
The latest generation M series sizes for extremely large databases or applications							
Most used by Azure users							
D2s_v3	General purpose	2	8	4	3200	16	Sup
D2as_v4	General purpose	2	8	4	3200	16	Sup
B2s	General purpose	2	4	4	1280	8	Sup
B2ms	General purpose	2	8	4	1920	16	Sup
DS2_v2	General purpose	2	7	8	6400	14	Sup
B4ms	General purpose	4	16	8	2880	32	Sup
D4s v3	General purpose	4	16	8	6400	32	Sup

Select | Prices presented are estimates in your local currency that include only Azure infrastructure costs and any discounts for the subscription and location. The prices don't include any applicable software costs. Final charges will appear in your local currency in cost analysis and billing views. View Azure pricing calculator.

Give feedback

2.10 Click Update

Update node pool ...

mycluster

Node size *

Standard D2s v3
2 vcpus, 8 GiB memory
[Choose a size](#)

Scale method

☒ Manual
☐ Autoscale - **Recommended**
 This option is recommended so that the cluster is automatically sized correctly for the current running workloads.

Node count *

1

Optional settings

Max pods per node *

110 30 - 250

Enable public IP per node

☐

Labels

Labels are key/value pairs that can be used to categorize or add identifying information to Kubernetes resources such as nodes. Labels for the node pool will be applied to each node in the node pool. [Learn more >](#)

Update **Cancel**

Update

2.11 Navigate to the Networking tab

Create Kubernetes cluster ...

Basics **Node pools** **Networking** Integrations Advanced Tags Review + create

Node pools

In addition to the required primary node pool configured on the Basics tab, you can also add optional node pools to handle a variety of workloads. [Learn more >](#)

[+](#) Add node pool [Delete](#)

☐	Name	Mode	Node size	OS SKU	Node count	Availat
☐	agentpool	System	Standard_D2s_v3	Ubuntu	1	1,2,3

Enable virtual nodes

Virtual nodes allow burstable scaling backed by serverless Azure Container Instances. [Learn more >](#)

Enable virtual nodes

☐

Previous **Next** **Review + create**

[Give feedback](#)

2.12 Under **Container networking**, provide the details as shown below:

Container networking

Network configuration ⓘ

☒ Kubenet
Best for smaller node pools. Each pod is assigned a logically different IP address from the subnet for simpler setup

☐ Azure CNI
Best for larger node pools. Each node and pod is assigned a unique IP for advanced configurations

Bring your own virtual network ⓘ

☐

DNS name prefix * ⓘ

mykcluster-dns

Network policy * ⓘ

☒ None
Allow all ingress and egress traffic to the pods
 ☐ Calico
Open-source networking solution. Best for large-scale deployments with strict security requirements
 ☐ Azure
Native networking solution. Best for simpler deployments with basic security and networking requirements

2.13 Navigate to the **Integrations** tab

Microsoft Azure

Search resources, services, and docs (G+)

odl_user_1155041@sim...
SIMPLILEARN HOL 100688 (SIMP...

Home > Kubernetes services >

Create Kubernetes cluster

Basics

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Review + create

Azure provides various networking controls to help manage and secure access to your Kubernetes cluster. [Learn more](#)

Private access

Enable a private cluster to restrict worker node to API access, enhancing your Kubernetes workload's security and isolation.

Enable private cluster ⓘ

☐

Public access

Set authorized IP ranges ⓘ

☐

Container networking

Network configuration ⓘ

☒ Kubenet
Best for smaller node pools. Each pod is assigned a logically different IP address from the subnet for simpler setup
 ☐ Azure CNI
Best for larger node pools. Each node and pod is assigned a unique IP

Previous

Next

Review + create

Give feedback

2.14 Under **Azure Monitor**, select **Off**

Azure Monitor

In addition to the CPU and memory metrics included in AKS by default, you can enable Container Insights for more comprehensive data on the overall performance and health of your cluster. Billing is based on data ingestion and retention settings.

[Learn more about container performance and health monitoring](#)

[Learn more about pricing](#)

☐ Default configuration
☐ Custom configuration
☒ Off

2.15 Click **Review + create**

Microsoft Azure

Search resources, services, and docs (G+)

Home > Kubernetes services >

Create Kubernetes cluster

Basics Node pools Networking Integrations Advanced Tags Review + create

Connect your AKS cluster with additional services.

Microsoft Defender for Cloud
Microsoft Defender for Cloud provides unified security management and advanced threat protection across hybrid cloud workloads. [Learn more](#)

✔ Your subscription is protected by Microsoft Defender for Cloud basic plan.

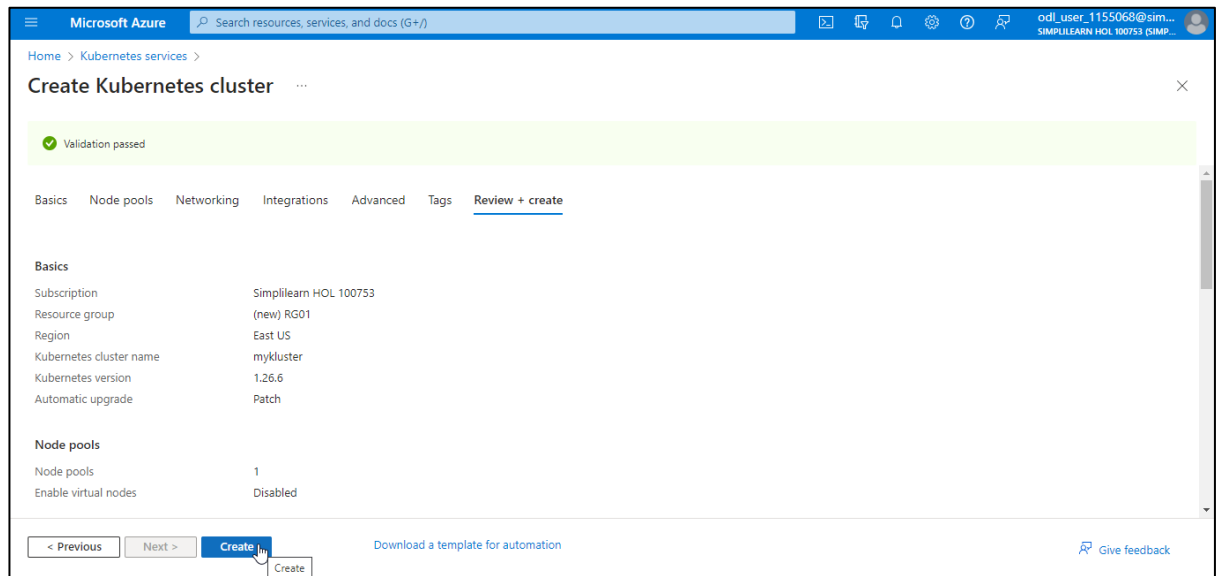
Azure Container Registry
Connect your cluster to an Azure Container Registry to enable seamless deployments from a private image registry. [Learn more](#)

Container registry: None
[Create new](#)

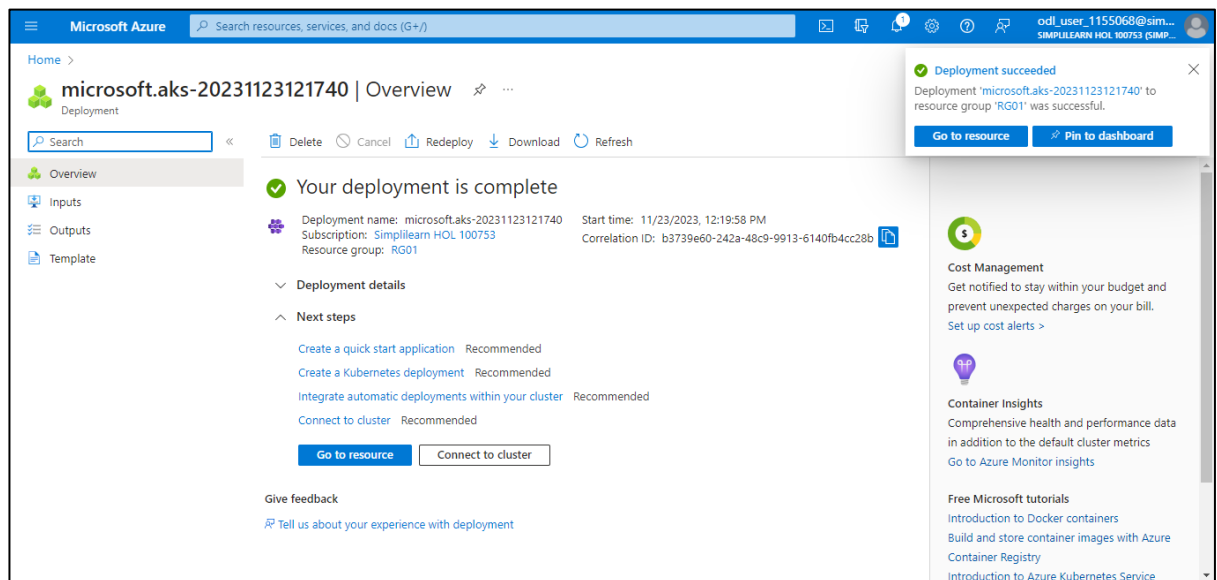
Azure Monitor
In addition to the CPU and memory metrics included in AKS by default, you can enable Container Insights for more comprehensive data on the overall performance and health of your cluster. Billing is based on data ingestion and retention settings. [Learn more about container performance and health monitoring](#)

[Give feedback](#)

2.16 Once the validation is passed, click **Create**

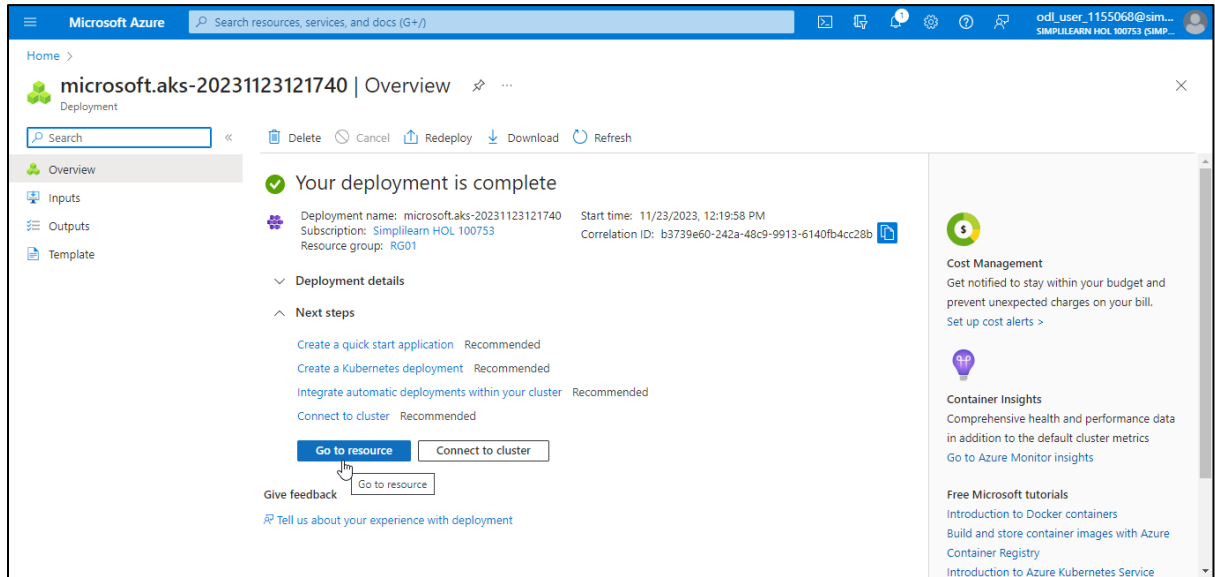


The deployment has been completed as shown in the screenshot below:

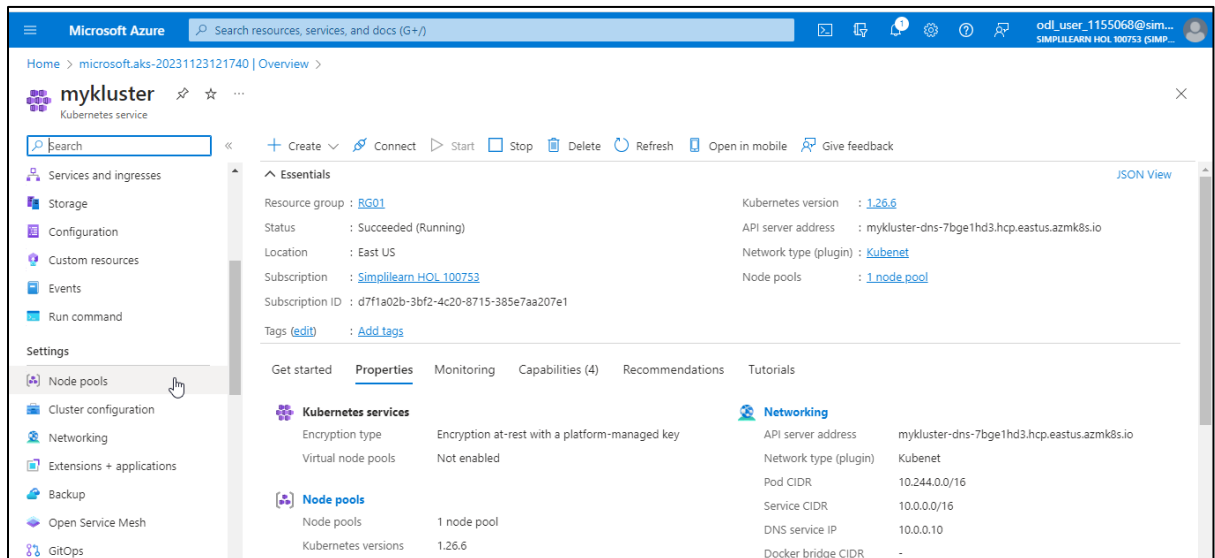


Step 3: Deploy pods into the Azure Kubernetes Service (AKS) cluster

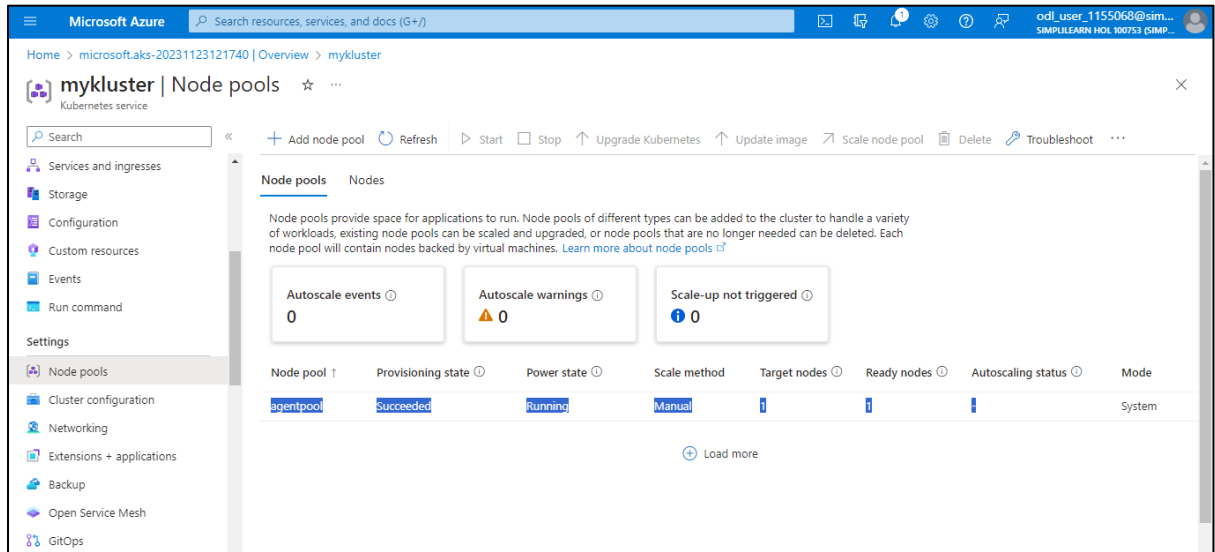
3.1 Click Go to resource



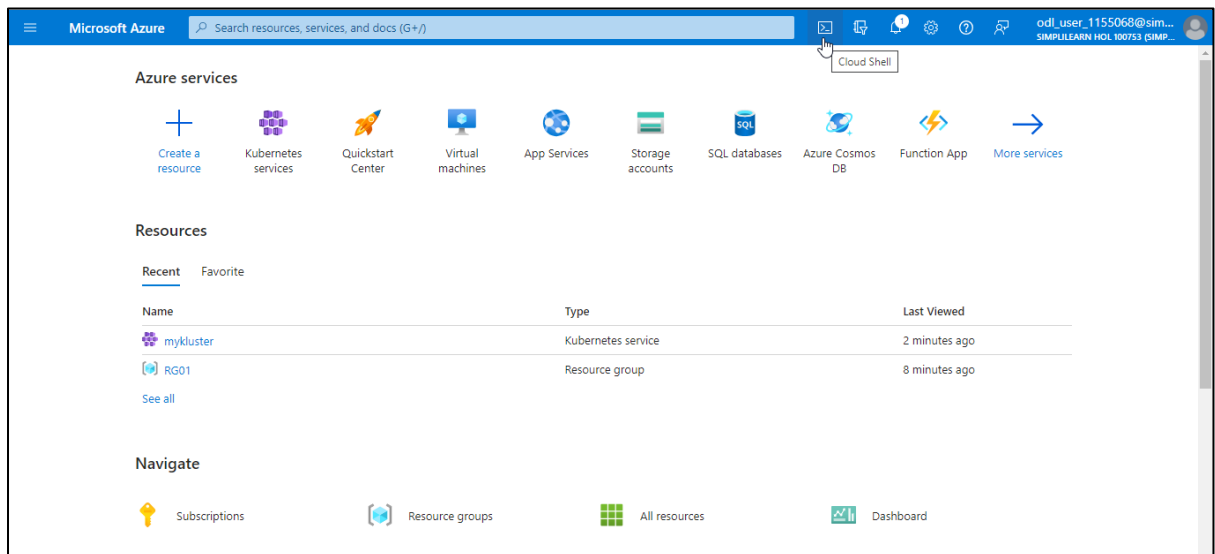
3.2 In the created cluster (mykluster) page, click Node pools under Settings



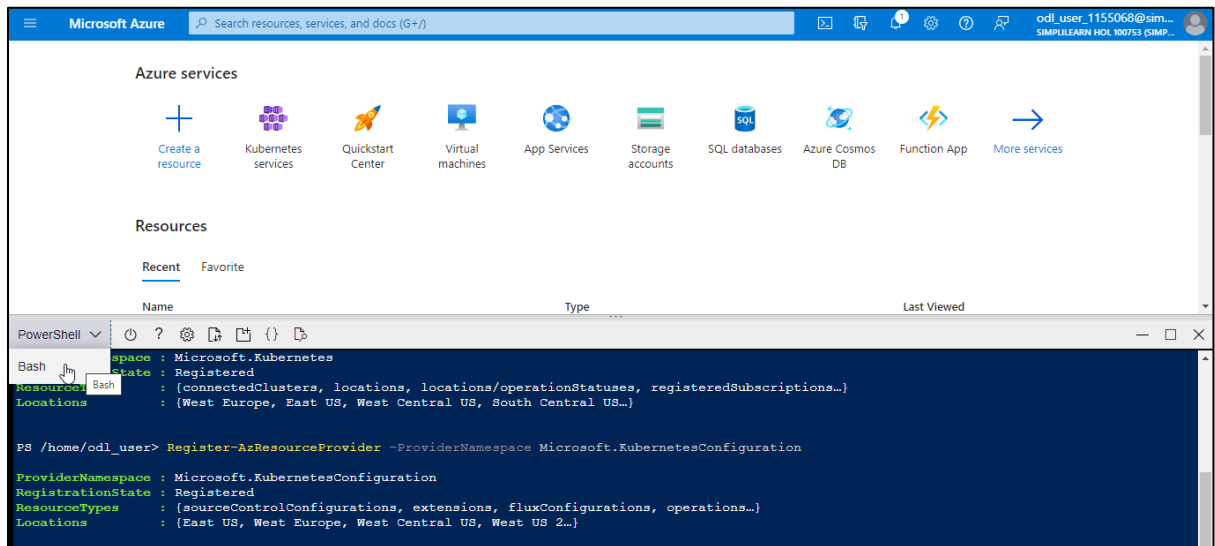
Verify that the cluster consists of a single node pool and is running successfully as shown below:



3.3 On the Microsoft Azure home page, click on Cloud Shell



3.4 Click on the **PowerShell** drop-down and select **Bash**

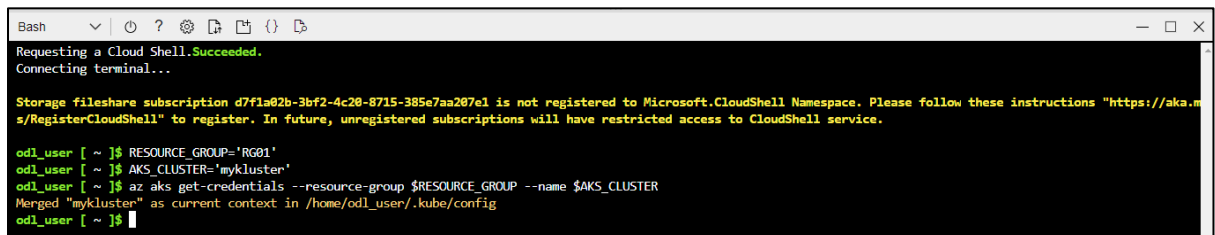


3.5 Run the following command to retrieve the credentials to access the AKS cluster:

RESOURCE_GROUP='RG01'

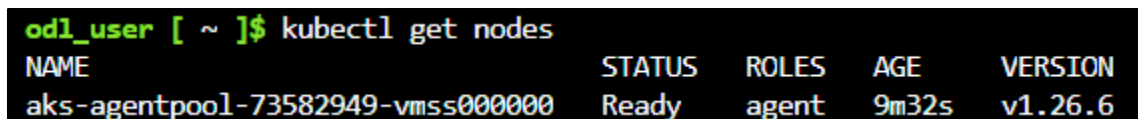
AKS_CLUSTER='mykluster'

az aks get-credentials --resource-group \$RESOURCE_GROUP --name \$AKS_CLUSTER



3.6 Run the following command to verify connectivity to the AKS cluster, and the **STATUS** should be **Ready**:

kubectl get nodes



3.7 Run the following command to deploy the **nginx** image from the Docker Hub:

kubectl create deployment nginx-deployment --image=nginx

```
odl_user [ ~ ]$ kubectl create deployment nginx-deployment --image=nginx
deployment.apps/nginx-deployment created
```

3.8 Run the following command to verify that a Kubernetes pod has been created:

kubectl get pods

```
odl_user [ ~ ]$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-55888b446c-8kfnw	1/1	Running	0	48s

3.9 Run the following command to identify the state of the deployment:

kubectl get deployment

```
odl_user [ ~ ]$ kubectl get deployment
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	1/1	1	1	94s

3.10 Run the following command to make the pod accessible from the internet:

kubectl expose deployment nginx-deployment --port=80 --type=LoadBalancer

```
odl_user [ ~ ]$ kubectl expose deployment nginx-deployment --port=80 --type=LoadBalancer
service/nginx-deployment exposed
```

3.11 Run the following command to identify whether a public IP address has been provisioned:

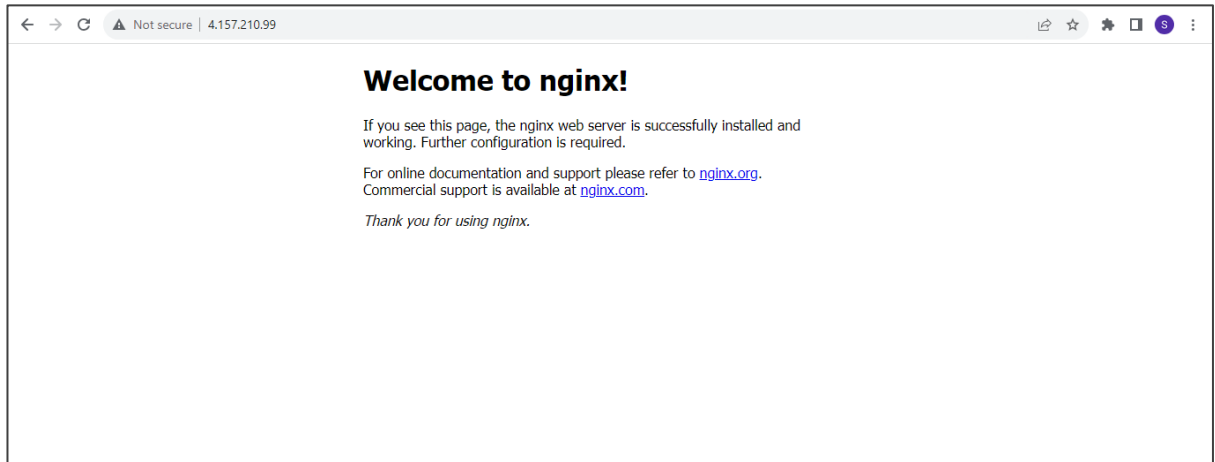
kubectl get service

```
odl_user [ ~ ]$ kubectl get service
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.0.0.1	<none>	443/TCP	17m
nginx-deployment	LoadBalancer	10.0.189.215	4.157.210.99	80:30788/TCP	3m7s

Note: Rerun the command until the value in the **EXTERNAL-IP** column for the nginx-\ deployment entry changes from **<pending>** to a public IP address. Note the public IP address in the **EXTERNAL-IP** column for nginx-deployment.

- 3.12 In the browser window, enter the IP address obtained from the previous step and verify that the browser page displays the **Welcome to nginx!** message:



Step 4: Scale containerized workloads in the Azure Kubernetes Service (AKS) cluster

- 4.1 Navigate back to the **Cloud Shell** pane and run the following command to scale the deployment by increasing the number of pods to 2:

```
RESOURCE_GROUP='RG01'
```

```
AKS_CLUSTER='mykluster'
```

```
kubectl scale --replicas=2 deployment/nginx-deployment
```

```
odl_user [ ~ ]$ RESOURCE_GROUP='RG01'
odl_user [ ~ ]$ AKS_CLUSTER='mykluster'
odl_user [ ~ ]$ kubectl scale --replicas=2 deployment/nginx-deployment
deployment.apps/nginx-deployment scaled
```

4.2 Run the following command to verify the outcome of scaling the deployment:

kubectl get pods

```
odl_user [ ~ ]$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-55888b446c-8kfnw	1/1	Running	0	12m
nginx-deployment-55888b446c-hjp82	1/1	Running	0	3m50s

4.3 Run the following command to scale the deployment:

kubectl scale --replicas=5 deployment/nginx-deployment

```
odl_user [ ~ ]$ kubectl scale --replicas=5 deployment/nginx-deployment
deployment.apps/nginx-deployment scaled
```

4.4 Run the following command to verify the outcome of scaling the deployment:

kubectl get pods

```
odl_user [ ~ ]$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-55888b446c-2lt6k	1/1	Running	0	41s
nginx-deployment-55888b446c-8kfnw	1/1	Running	0	14m
nginx-deployment-55888b446c-9ljrr	1/1	Running	0	41s
nginx-deployment-55888b446c-hjp82	1/1	Running	0	5m29s
nginx-deployment-55888b446c-jf6ml	1/1	Running	0	41s

4.5 Run the following command to review the pod distribution across cluster nodes:

kubectl get pod -o=custom-columns=NODE:.spec.nodeName,POD:.metadata.name

```
odl_user [ ~ ]$ kubectl get pod -o=custom-columns=NODE:.spec.nodeName,POD:.metadata.name
```

NODE	POD
aks-agentpool-73582949-vmss000000	nginx-deployment-55888b446c-2lt6k
aks-agentpool-73582949-vmss000000	nginx-deployment-55888b446c-8kfnw
aks-agentpool-73582949-vmss000000	nginx-deployment-55888b446c-9ljrr
aks-agentpool-73582949-vmss000000	nginx-deployment-55888b446c-hjp82
aks-agentpool-73582949-vmss000000	nginx-deployment-55888b446c-jf6ml

4.6 Run the following command to delete the deployment:

kubectl delete deployment nginx-deployment

```
odl_user [ ~ ]$ kubectl delete deployment nginx-deployment  
deployment.apps "nginx-deployment" deleted
```

By following these steps, you have successfully deployed an Azure Kubernetes Service (AKS) cluster, deployed pods into the Azure Kubernetes Service (AKS) cluster and scaled containerized workloads in it.